MIL-T-22647B(WEP)
3 April 1963
SUPERSEDING
MIL-T-22647A(WEP)
1 June 1962

MILITARY SPECIFICATION

TEST SET, COMPUTER TS-1393A/APQ-83

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be capable of testing the functional operation and accuracy of the CP-473/APQ-83 and the CP-473A/APQ-83 Missile Release Computers. The unit shall contain the necessary circuits to supply primary power, functional controls, simulated data inputs and readout devices for the computers circuits.

1.2 Classification - Test Set, Computer TS-1383A/APQ-83 shall be of one type and shall consist of the following items:

Item	Type Designation	Appl. Para.
Test Set, Computer	TS-1393A/APQ-83	3.3.1
Wiring Harness, Branched	W1 of TS-1393/APQ-83	3.3.2
Wiring Harness, Branched	W2 of TS-1393/APQ-83	3.3.3
Wiring Harness	W3 of TS-1393/APQ-83	3.3.4
Cover, Test Set		3.2.4.1

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-R-21642	Radar Set AN/APQ-83, Military Specification for
MIL-R-23373	Radar Set AN/APQ-94, Military Specification for
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment

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MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-D-18300	Design Data Requirements for Contracts Covering Airborne Avionic Equipment
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment, for Use with Electronic and Fire Control Systems, General Specification for

PUBLICATIONS

SAR-300	Material Changes and Material Bulletins; Preparation of

Federal

TT-E-527	Enamel, Alkyd, Lustreless

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

2.3 Precedence of Documents - When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

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3.1.1 Non-Standard Parts and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682.

3.2 Design and Construction -

3.2.1 General - This equipment shall fulfill the applicable design, workmanship and construction requirements of Specification MIL-T-21200 for Class 3 equipment unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-T-21200 for Class 3 equipment, except as modified herein.

3.2.2.1 Vibration - In accordance with Specification MIL-T-21200 paragraph 3.2.14.4 with these exceptions:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 33	0.02 inch

A test at 0.02 inch double amplitude with the input frequency varying between 33 and 55 cycles per second is to be run for exploratory determination of points of high mechanical transmissibility. This test is to be non-destructive.

3.2.2.2 Sand, Dust, Fungus and Explosion - Sand, dust, fungus and explosion requirements of Specification MIL-T-21200 shall not be applicable to this equipment.

3.2.3 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment.

Temperature	Room Ambient
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90% relative humidity
Input Voltage	115 ± 1.0 VAC and 27.0 ± 0.5 VDC

3.2.4 Form Factor - The test set shall be housed in a combination case fulfilling the requirements of MIL-T-21200. The overall dimensions of the test set with cover installed shall not exceed 13.5 inches wide by 14.5 inches high by 11 5 inches deep. The total weight of the equipment, excluding cables, shall not exceed 33 pounds.

3.2.4.1 Cover, Test Set - The combination case shall be supplied with a detachable cover in accordance with Specification MIL-T-21200. The cover shall contain provisions for stowing an instruction book, the cable assemblies, allen wrenches and spare fuses.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 24 hours, continuously or intermittently, without the necessity for readjustment of any controls.

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3.2.6 Operating Life -

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.7 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-T-21200, Class 3, and shall give specified performance from the following power sources with characteristics as defined herein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase)

115 Volt Constant Frequency (Type 1)

Operating Voltage Limits	105 to 127 volts
Operating Frequency Limits	380 to 420 cps
Power (1 ϕ)	300 VA max

(2) DC Power 28 Volts

Operating Voltage Limits	25 to 31 volts
Current	0.6 amp max

3.2.10 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other devices shall be held to a minimum.

3.2.10.1 Undervoltage Protection - The equipment shall not be damaged by operation at 98 VAC and/or 21 VDC, and shall automatically resume normal operation when the voltage returns within limits.

3.2.11 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.12 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.13 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

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3.3 Detail Requirements -

3.3.1 Test Set, Computer TS-1393A/APQ-83 -

3.3.1.1 Functions - The test set shall be capable of testing the operation and accuracy of Missile Release Computer CP-473/APQ-83 or Missile Release Computer CP-473A/APQ-83.

3.3.1.2 Finish - The case shall be finished with gray enamel conforming to Specification MIL-T-21200. The panel shall be finished with black synthetic lustreless enamel in accordance with TT-E-527.

3.3.1.3 Controls and Indicators - The following controls and indicators, labeled as specified, shall be provided on the front panel of the Test Set Computer TS-1393A/APQ-83.

(1) POWER/OFF/RESET - A main power switch shall be provided on the front panel to connect or disconnect the incoming primary power. A momentary position of the switch shall enable recycling of the power supply overload relay.

(2) MISSILE SELECTOR - A three position toggle switch shall be provided to control the missile identification circuits of the computer under test per paragraph 3.3.1.4.10.

(3) LINE ADJUST - A control shall be provided to enable adjustment of the primary AC power to the computer under test and the test set power supply.

(4) LINE VOLTAGE - An expanded scale AC voltmeter shall be provided to indicate the output of the LINE ADJUST control. The meter shall be accurate to within ± 2%.

(5) POWER lamp - A panel lamp shall be provided and shall illuminate when power is applied to the unit by the POWER/OFF/RESET switch and shall be red in color.

(6) COINCIDENCE TEST lamp - A panel lamp shall be provided and shall illuminate to indicate operation of either the computer coincidence or ARO relay circuitry as determined by the position of the TEST SELECTOR switch.

(7) POWER SUPPLY ADJUST - Provision shall be made to enable front panel adjustment of the +150 VDC reference diode current.

(8) TEST SELECTOR - A ten position rotary switch shall be supplied to select type of test to be performed by the test set.

(9) COMPUTER PROBLEMS - A ten position rotary switch shall be provided to control type and parameters of computer problems.

(10) MAG-AMP CURRENT TEST - A seven position rotary switch shall be provided for selecting the computer mag-amp circuit to be monitored.

(11) METER NO. 1 and METER NO. 2 - Two -50/0/+50 microammeter panel meters shall be provided to function as multimeters. The function of these meters shall be controlled by the TEST SELECTOR switch.

(12) COINCIDENCE TEST ADJUST - A ten turn potentiometer equipped with an indicating dial shall be provided to act as a source of variable voltage between 0 ± 0.1 and +27 ± 2.0 VDC. The function of this control shall be controlled by the TEST SELECTOR switch.

(13) METER NO. 1 and METER NO. 2 SENSITIVITY - Two four position rotary switches equipped with a spring return to the maximum CCW position shall be

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provided to control the sensitivity of METER NO. 2. Full scale meter sensitivity of ±50 V ± 5%, ±5 V ± 5%, ±0.5 V ± 5% and ±0.05 V ± 5% shall be selectable by these controls.

(14) RANGE POTENTIOMETER - A ten turn potentiometer equipped with an indicating dial shall be provided to act as a source of variable radar range voltage between 0 ± 0.1 VDC and +150 ± 5 VDC.

(15) REFERENCE POTENTIOMETER - A readout device shall be provided which, when used to null METER NO. 2, shall indicate the value of voltages between zero and +50.0 VDC and shall consist of the following controls:

(a) COARSE - A five position switch shall be provided and shall indicate the tens digit of the unknown voltage.

(b) FINE - A ten turn potentiometer equipped with an indicating dial shall be provided. The three dial numerals shall indicate the units, tenths and hundredths digits of the unknown voltage when METER NO. 2 is adjusted to read a null.

3.3.1.4 Computer Power Supply - When supplied with primary power as specified in paragraph 3.2.9, the test set shall provide the following primary power voltages to the computer under test:

(1) 115 ± 2% VAC 380 to 420 cps
(2) +27 ± 2 VDC
(3) +150 ± 5 VDC

3.3.1.4.1 Reference Power Supply - When supplied with primary power as specified in paragraph 3.2.9, the test set power supply shall provide the following reference voltages:

(1) +150 ± 5 VDC
(2) +100 ± 1% VDC
(3) +20.0 ± 0.04 VDC
(4) +13.748 ± 0.028 VDC
(5) +10.927 ± 0.022 VDC
(6) +9.276 ± 0.018 VDC
(7) -1.00 ± 0.005 VDC
(8) -5.616 ± 0.011 VDC
(9) -9.572 ± 0.019 VDC
(10) -16.029 ± 0.032 VDC
(11) -20.00 ± 0.040 VDC

3.3.1.4.1.1 Reference Power Supply - Overload Protection - An overload relay shall be included to disconnect the primary power if the loading on the power supply becomes excessive. A means of resetting the overload relay shall be provided on the front panel. Operation of the overload relay shall be indicated by the loss of a voltage indication on the LINE VOLTAGE meter.

3.3.1.4.2 Reference Potentiometer - The test set shall contain a bridge voltmeter capable of indicating potentials between zero and +50.00 VDC with an accuracy of 0.01 V or 0.1% of the measured value whichever is greater. The value of the unknown voltage shall be indicated in increments of 0.01 volts on direct reading front panel controls when balance is indicated by a panel meter.

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3.3.1.4.3 Range Volts Scaled Test - The test set shall be capable of measuring with an accuracy of 0.1% the ratio of the test set simulated range voltage input to the computer versus the computers range voltage scaling circuitry output with an accuracy of 0.1%.

3.3.1.4.4 Range Potentiometer - The test set shall be capable of simulating a radar range voltage input to the computer under test. The radar range voltage shall be adjustable between 0 ± 0.01 V and +150 ± 5 VDC by means of a front panel control.

3.3.1.4.5 Range Volts Scaled Absolute Value Test - The test set shall be capable of measuring the value of the output of the computer Range Volts Scaled circuitry with an accuracy of 0.1%.

3.3.1.4.6 Computer Power Supply Test - The test set shall contain the circuitry required to measure the output of the computer -20 VDC and the +20 VDC regulated power supplies. These voltages shall be measured with an accuracy of ± 0.04 V at ± 20 VDC.

3.3.1.4.7 Air Speed Calibration - The test set shall contain the circuits necessary to indicate that the voltage output of the computer air speed (P_t) transducer has been adjusted to any one of the three following values:

(1) +9.276 ± 0.018 VDC

(2) +10.927 ± 0.022 VDC

(3) +13.748 ± 0.028 VDC

3.3.1.4.8 Altitude Calibration - The test set shall contain the circuits necessary to indicate when the voltage output of the computer Altitude $[F_2 (PS)]$ transducer has been adjusted to any one of the following values:

(1) -5.616 ± 0.011 VDC

(2) -9.572 ± 0.019 VDC

(3) -16.029 ± 0.032 VDC

3.3.1.4.9 Simulated Range Rate - The test set shall provide a simulated range rate voltage of either 0 ± 0.01 VDC or -1.00 ± 0.01 VDC to the computer.

3.3.1.4.10 Missile Identification - The test set shall contain the circuits necessary to operate the computer missile identification relays in the following modes:

(1) 1A

(2) MK-29

(3) MK-30

3.3.1.4.11 Missile Rmax Test - The test set shall contain the circuits necessary to simulate or control computer inputs of radar range, radar range rate, altitude, airspeed and missile identification for three missile Rmax problems. The computers solution to the problems shall be indicated by the Reference Potentiometer on the test set.

3.3.1.4.12 Missile Rmin Test - The test set shall contain the circuits necessary to simulate or control computer inputs of radar range, radar range rate, altitude, airspeed and missile identification for a single fixed Rmin problem. The computers solution to the problems shall be indicated by the Reference Potentiometer on the test set.

3.3.1.4.13 Missile Amax Test - The test set shall contain the circuits necessary to simulate or control computer inputs of radar range, radar range rate, altitude, airspeed and missile identification for two fixed Missile Amax problems. The computers solution to the problems shall be indicated by the Reference Potentiometer on the test set.

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3.3.1.4.14 Missile Dmax/R Test - The test set shall contain the circuits necessary to simulat or control computer inputs of radar range, radar range rate, altitude, airspeed and missile identification for two fixed Missile Dmax/R problems. The computers solution to the problems shall be indicated by the Reference Potentiometer on the test set.

3.3.1.4.15 Computer Mag Amp Current Test - The unit shall be capable of monitoring the control winding current of each of the six CP-473A/APQ-83 computer magnetic amplifiers. The amplitude and polarity of the current shall be indicated by METER NO. 1.

3.3.1.4.16 Coincidence Circuit Test - The test set shall contain the circuits required to contr the voltage inputs and monitor the voltage differential between the inputs to each of the computers Missile Rmax, Missile Rmin and Missile Amax coincidence relay circuits. A panel lamp shall be provided to indicate the coincidence circuits on or off condition.

3.3.1.4.17 ARO Relay Test - The test set shall provide the necessary controls to operate the ARO relay in the computer. A panel lamp shall be provided to indicate the relays on or off condition.

3.3.1.4.18 Range Volts Limited Test - The test set shall contain the necessary circuits to tes the condition of the Computers Range Volts Limited circuitry.

3.3.1.4.19 Resistor Module Substitution - The test set shall contain one resistor module Z3305-IC (Magnavox part number 712586-801), one resistor module Z3304-IC (Magnavox part number 712585-801), and the required circuitry and patch cables to enable direct substitution for the corresponding modules in the computer.

3.3.1.5 Connectors - The following external connectors shall be provided:

(1) Power-Computer Control (J501)

(2) Computer Test Signals (J502)

(3) Resistor Module Test (J503)

3.3.2 Wiring Harness, Branched W1 of TS-1393A/APQ-83 - The equipment shall be furnished with a 3-foot power cable to provide power connection to the test set and to the computer unde test. One end of the cable shall be terminated with a connector for connection to the test set. The branched ends of the cable shall be terminated with the following connections:

(1) A connector shall be provided for connection to an external +27 VDC power source.

(2) An MS3107A-20-4P connector shall be provided for connection to an external 115 V 400 cps power source.

(3) A connector shall be provided for connection to the computer under test.

3.3.3 Wiring Harness, Branched W2 of TS-1393A/APQ-83 - The equipment shall be furnished with a 3-foot cable to provide connection between the test set and the computer under test. One end of the cable shall be terminated with a connector for connection to the test set. The branched end of the cable shall be terminated with two connectors.

3.3.4 Wiring Harness, Branched W3 of TS-1393A/APQ-83 - The equipment shall be furnished with a 3-foot cable to provide connection between the test set and the computer under test. One end of the cable shall be equipped with connector for connection to the test set. The branched ends shall provide two connectors for connection to the computer under test.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances of values setting forth acceptable variations from the performance under the standard conditions will be specified.

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3.5 Preproduction Sample - Unless otherwise specified in the contract or order, a preproduction sample of the Test Set, Computer TS-1393A/APQ-83 shall be manufactured such that it is representative of the production lot in all respects. The sample will be tested as specified in section 4 herein and is for the purpose of determining that the contractor prior to starting his production is capable of yielding items that comply with the technical requirements of the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test Required - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests
(2) Sample Tests
(3) Acceptance Test
(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests
(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration test shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with MIL-T-5422 to the extent specified herein, interference tests in accordance with MIL-I-6181 and drop tests for transit and combination cases in accordance with Specification MIL-T-21200.

4.2.3 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract.

4.3 Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and shall have successfully passed the individual tests.

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4.3.1 Scope of Test - As a minimum, each equipment selected for sampling tests shall be subjected to the following test:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

4.3.2 Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Special Tests

(3) Life Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated at the conditions specified herein a total accrued running time of 6 hours. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a maximum double amplitude of 0.02 inches.

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The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes as specified herein. The direction of vibration should be vertical to the normal mounting plane. The equipment shall be non-operating during the 10 minute vibration period. The 6-hour period of operation shall include that time necessary for performing all equipment tests and calibrations. Only that time necessary to make a total accrued running time of 6 hours is required for operation after the 10-minute vibration period. During the 6-hour period of operation the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test continued. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures.

4.4.2 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.2.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require variations to assure satisfactory control.

4.4.2.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minute cycling in each of 3 planes)

(3) Humidity Tests

4.4.3 Life Tests - When requested by the procuring activity, one unit shall be selected from current production equipment and shall have successfully passed the Individual Tests.

4.4.3.1 Scope of Test - A 200 hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.4.3.1.1. A performance check shall be made to determine the equipment is operating satisfactorily.

4.4.3.1.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal Room
Altitude	Normal Ground (0 - 5000 ft.)
Humidity	Room Ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	27.0 ± 2.0 volts

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4.4.3.1.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3.1.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subje to approval by the procuring activity.

4.4.3.2 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A reco shall be kept of all failures throughout the test, including all tube failures. This record shall indicat the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.4.3.3 Failure Report - In the event of a failure, the government inspector shall be notif immediately. A report shall be submitted to the procuring activity upon completion of the test. In t report, the contractor shall propose suitable and adequate design or material corrections for all fail ures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acce ance tests and life tests shall be prepared by the contractor and submitted to the procuring activity f review and approval. The right is reserved by the procuring activity or the government inspector t modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a gu for preparation of test procedures. When test procedures are available from previous contracts su procedures may be used when approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitte by the contractor until it has been previously tested and inspected by the contractor and found to con ply, to the best of his knowledge and belief, with all applicable requirements.

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4.7 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes, and Material Bulletins shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.0 ± 0.5 VDC and 115 ± 1.0 VAC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 49-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION (No. and abbrev. title)

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? *(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)*

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY (Print name and title)	DATE

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OFFICIAL BUSINESS

Chief, Bureau of Naval Weapons (RREN-5)

Navy Department

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